Enhancing Classroom Accessibility for Deaf Students through Sign Language Recognition

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***Abstract*—This paper addresses the communication barriers faced by the deaf and voiceless community by introducing a system that translates sign language into spoken or written language. The primary aim is to foster better communication between these communities and the hearing population, ultimately supporting academic success for individuals with hearing impairments. By utilizing the deep learning model VGG19, the system is able to recognize sign language with an astounding 99.97% accuracy rate, which is important for guaranteeing fair access to education. Because VGG19 can efficiently extract complicated features from images, a property that is essential for precisely interpreting the intricate hand gestures and motions that are a part of sign language communication, it is used in sign language recognition. It highlights the importance of real-time captions and interpretations as transformative tools in educational environments. Furthermore, the paper explores the implementation of this technology, outlining its benefits for educators and deaf students alike. Through examination of challenges, and future directions, it underscores the growing significance of sign language recognition in education, emphasizing its role in promoting inclusivity and enhancing learning experiences.**

Keywords—Sign language recognition, Deep Learning, VGG19, Communication accessibility, Academic success, Deaf education, Real-time interpretation

I. INTRODUCTION

Through hand gestures, facial expressions, and body language, deaf and hearing-impaired individuals who use sign language as their primary means of communication can convey ideas, feelings, and thoughts. Effective communication and inclusivity for this population are greatly aided by the understanding and interpretation of sign language motions. Convolutional Neural Networks (CNNs) have demonstrated exceptional efficacy in a range of visual identification tasks, such as object detection, image categorization, face expression recognition, and more [1].

This work aims to give an assessment of the current state-of-the-art in CNN-based sign language recognition. In recent years, there has been a significant increase in the study of computer vision for sign language recognition (SLR). Among the primary problems in SLR are feature representation, sign classification, and sign camera background modeling. The ability to smoothly and accurately recognize sign language gestures is essential for the effective communication of the hard of hearing and deaf. The incorporation of deep learning algorithms has transformed many fields as technology develops quickly. The main goal is to develop effective methods that can quickly identify and categorize different kinds of sign language motions [2].

The field of sign language recognition has seen the use of numerous algorithms and techniques. Because Convolutional Neural Networks (CNNs) are so good at image classification tasks—including identifying sign language gestures—they have become industry standard technology. Recurrent neural networks (RNNs) include Long Short-Term Memory (LSTM) networks, which are highly useful for recognizing the temporal connections present in sequential data. This is important because it helps to comprehend how sign language expressions change over time. Modeling the changes between various states or postures in sign language movements is a useful application for Hidden Markov Models (HMMs), which are well-known for their skill in sequential data processing. One further classification system that may be useful in identifying particular signs or movements in sign language is Support Vector Machines (SVMs).

Additionally, other deep learning architectures outside of CNNs and LSTMs, like Transformer models, Recurrent Convolutional Neural Networks (RCNNs), and Residual Networks (ResNets), provide better scalability and performance for a variety of sign language identification tasks. The choice of algorithm is influenced by various aspects, including the application requirements, processing resources, dataset size, and gesture complexity. In order to meet the varied demands of the community and promote inclusion in communication and education, sign language recognition technology is continually evolving due to ongoing study and innovation in algorithm development.

Accurate categorisation is essential for creating inclusivity and removing barriers to communication. To accomplish our first goal, this makes use of state-of-the-art deep learning techniques, including Convolutional Neural Networks (CNN) and VGG19, which are well-known for their proficiency in image identification tasks. CNNs are very good at extracting features from images, which makes them ideal for deciphering the intricate visual cues included in sign language gestures [3]. On the other hand, the complexity and depth of the VGG19 architecture improve our classification abilities and lay the groundwork for a sound and dependable model. A major improvement in accessibility has occurred with the transition from anticipated sign language movements to audio output. It guarantees that people with hearing impairments can not only comprehend but also actively participate in the

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messages that are being given, furthering a sense of inclusion and understanding [4]. The deep learning models are trained and validated using a large and varied dataset. The value of inclusive data that takes into account different sign languages, regional differences, and dialects.

II. Background

Ensuring that deaf children have equal access to literacy and knowledge is vital for their academic and social development. To create a conducive learning environment for them, teachers should focus on deaf-friendly strategies. Deaf students may have diverse communication methods, including sign language for those who are prelingually deaf, and some may use cochlear implants or hearing aids. Others might combine spoken language with sign language, while some rely on lip reading.

The lack of accessible auditory information remains a significant obstacle, posing challenges related to comprehension, information access, and communication for these children. A study conducted in Jordan highlighted the multifaceted difficulties faced by students with hearing loss, encompassing communication, academics, social interactions, emotions, and family dynamics. The significance of customized language, speech, and audiological rehabilitation programs for students who are deaf or hard of hearing is highlighted by this study. Although the government provides essential support to help these students reach their full potential and ensure their optimal development, there remains a need for additional facilities and improvements [5].

A. Sign Language as A Bridge

For many deaf people, sign language serves as their main form of communication, and it has enormous value in educational settings. Sign language has its own syntax, grammar, and vocabulary. For many deaf individuals, it is their first and most accessible language. Just as spoken language that is integral to the hearing population, sign language is crucial for deaf individuals to express themselves, convey complex ideas, and engage in meaningful communication. They evolve naturally within deaf communities, reflecting the culture and identity of those who use them. Sign language provides a way for deaf individuals to access information and communicate with others. It bridges the communication gap that exists between deaf and hearing individuals, enabling them to interact in various settings, including education, employment, healthcare, and social interactions [6].

Deaf individuals can convey their emotions through facial expressions, body movements, and signs. It also empowers the deaf individuals to be more independent in their daily lives and enables them to make choices, express their needs and preferences, and navigate various situations. Sign language is essential for deaf children's educational development. It allows them to access curriculum content, engage with teachers and peers, and develop language skills necessary for cognitive and social development. Many deaf individuals attend specialized schools where sign language is the primary medium of instruction. Proficiency in sign language opens up career opportunities in fields such as interpreting, education, and advocacy. Deaf individuals who are fluent in sign language can pursue careers as sign language interpreters, educators, or advocates for the deaf community.

B. Rise of Sign Language Recognition

For those who are deaf or hard of hearing, sign language (SL) is the primary means of communication. It consists of both non-manual cues like facial expressions and manual gestures like hand and body movements. Expressions that communicate the meanings of words or phrases are made by combining and connecting these components.

The Deaf community places great emphasis on comprehending and recording these expressions and words. The need for creating sign language tools to help deaf people communicate and be included in society has long been acknowledged by the scholarly community. The vast range of sign languages and the extensive annotated datasets provide challenges, but recent advances in machine learning and artificial intelligence (AI) have been essential to automating and advancing these technologies [7].

The resolution and discriminating capabilities of sensors, together with the difficulties presented by occlusions and fast hand motions, are currently the main limitations on the accuracy of sign language capture technologies. The development of machine learning algorithms that can precisely classify human articulations as single signs or lengthy sentences is necessary for the process of sign language recognition (SLR). The absence of large annotated datasets, which has an impact on the accuracy and generalizability of SLR techniques, is one of the present limitations of SLR. Success in machine learning algorithms responsible for reliable sign recognition depends on the precise extraction of hand and body movements, along with facial expressions [8].

III. Sign Language Recognition Technology

A. Overview

In order to identify and comprehend sign language motions, sign language recognition technology uses computer vision and machine learning algorithms. This makes communication more efficient and inclusive. Sign language recognition begins with the detection of sign language gestures [9].

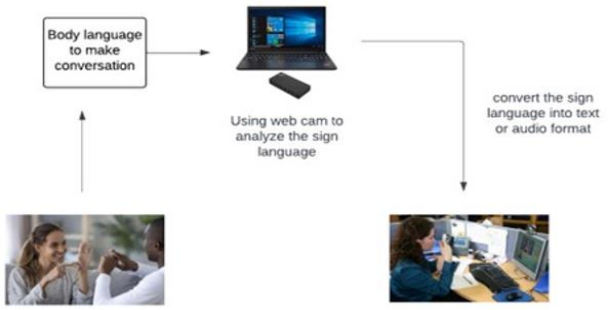


Fig. 1. Sign Language Architecture.

Fig 1 shows the technical structure of systems that automatically recognize and translate sign language into spoken language or text. These systems typically follow a pipeline involving:

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1) Sign Detection: Employing sensors or cameras to record the sign language movements.

2) Feature Extraction: Analysing the hand positions, movements, and other visual cues from the captured signal.

3) Recognition: Utilizing machine learning techniques, such as neural networks, to categorize the elements that were retrieved and determine the associated sign.

4) Translation: Converting the recognized sign into spoken language or text.

This can be achieved through different types of sensors, including cameras, gloves with built-in sensors, or depth-sensing devices like Microsoft Kinect. These sensors capture the movements and positions of the signer's hands, facial expressions, and body gestures. The system collects data on sign language gestures, creating a database of signs and their corresponding meanings. In order to train machine learning models, this data is necessary. The data is processed using machine learning methods, namely deep learning neural networks, which are used to identify sign language gestures. The data is processed using recurrent neural networks (RNNs) and convolutional neural networks (CNNs). The recorded data is subjected to feature extraction algorithms in order to emphasize important aspects of sign language motions, like hand shape, motion trajectory, and face expressions. Once the system has learned to recognize sign language gestures, it translates them into text or speech [10].

B. Recognition Techniques

1) Real-Time: Real-time captions involve the conversion of spoken language into text, which is displayed on a screen or device in real-time. Captions are typically generated by a trained captioner or using automatic speech recognition (ASR) technology. Real-time captions are suitable for a broad range of situations, including media and virtual meetings, providing universal accessibility [11].

2) Sign Language Interpretation: Sign language interpretation involves a trained interpreter who translates spoken language into sign language using gestures, facial expressions, and body movements. They are trained not only in language but also in the cultural nuances of the Deaf community, promoting effective communication. Sign language is a two-way communication method that may be used for interactive conversations and dialogues since it can express not just words but also emotions and tones [12].

IV. Implementation In Educational Settings

A. Data Acquisition

Collect a dataset of sign language gestures or signs. This dataset should include video or image sequences of people signing various words or phrases in sign language. It's important to have a diverse dataset with different signers, lighting conditions, and backgrounds.

B. Recognition Process

Preprocess the collected data to extract relevant features. This involves process such as video or image normalization, removal of background, and tracking of hand to isolate the signing hand or hands. Extract meaningful features from the pre-processed data. Common features for sign language recognition include hand shape, hand motion, finger positions, and facial expressions. These characteristics are necessary for machine learning model training. More complex models for sign language recognition can be created using Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs). For instance, you can use 2D CNNs for image-based sign recognition and RNNs for sequences of signs. Implement the recognition logic that takes the output of the machine learning model and maps it to the corresponding sign or word in the sign language. This may involve post-processing and context-aware algorithms [13].

C. Display and Delivery

Make the system's interface easy to use so that users can communicate with it. This interface can be in the form of a mobile app, desktop application, or even a hardware device like a glove or camera-based system [14].

D. Benefits and Impact

Sign language recognition technology plays an effective role in promoting inclusive learning environments. By accurately translating sign language into written or spoken language, it breaks down communication barriers for deaf students, ensuring they have equal access to educational content

An important benefit of sign language recognition technology is that it facilitates better communication between teachers and students who are deaf. Educators can effectively convey information through sign language, which the technology then translates in real-time [15].

The integration of sign language recognition technology creates a level playing field for deaf students in mainstream education. It ensures they can actively participate in class discussions, ask questions, and express their thoughts and ideas without any communication barriers. This technology, by providing real-time translation, helps educators understand and respond to the needs of deaf students promptly

E. Algorithm

Deep learning tasks are the main application for the VGG19 algorithm, a model that uses picture classification. While it's a powerful architecture for recognizing objects in images, it may not be the most suitable choice for sign language recognition, which involves video data and sequential patterns of hand movements and gestures. Recurrent neural networks (RNNs), convolutional neural networks (CNNs), or hybrid models that can analyse sequential input and capture temporal dependencies are usually needed for sign language recognition.

A model built for video data or sequences, such as a 3D CNN (Convolutional Neural Network), an LSTM (Long Short-Term Memory) network, or a combination of the two, may be used for sign language recognition. These models can effectively capture the dynamic nature of sign language gestures and recognize signs based on the order and motion of hand and body movements. While VGG19 is not directly suited for sign language recognition, you can adapt its principles for feature extraction from individual frames within a sign language video and then combine these features with a suitable sequence modelling network to

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build a more appropriate system for sign language recognition.



Fig. 2. Sign Language Recognition System.

Fig 2 the diagram outlines a system that translates sign language gestures into text. It is divided into four main stages:

1) Input: This stage captures the sign language gestures using a camera or sensor.

2) Feature extraction: The system extracts key features from the captured video, such as hand shapes, movements, and relative positions.

3) Recognition: A machine learning model analyses the extracted features and classifies them into the corresponding signs.

4) Output: The recognized signs are converted into text and displayed on a screen or sent to another device.

It also includes a feedback loop, suggesting the system can learn and improve its recognition accuracy over time. The "database" refers to a collection of sign language data used to train the machine learning model. The "processing" might involve additional steps like noise reduction or hand segmentation before feature extraction.

F. VGG19 Architecture

1. The VGG network takes an RGB image with a fixed size of 224x224 pixels, represented as a (224, 224, 3) matrix.
2. Each pixel is preprocessed by deducting from it the mean RGB value computed throughout the training set.
3. It uses 3x3 kernel sizes with a stride of 1 pixel to cover the entire image while maintaining spatial resolution through padding.
4. The Rectified Linear Unit (ReLU) activation is used to increase classification accuracy and introduce non-linearity.
5. The architecture includes three fully connected layers, with the first two having 4096 units and the third being a softmax layer for 1000-way ILSVRC classification.
6. A Model Checkpoint callback function is used to save the model's weights to disk at the end of each training epoch, ensuring that the best model is saved based on validation loss improvements.
7. The callback monitors 'val_loss' and saves weights only when there is an improvement in validation loss.
8. The model is trained for ten epochs using the Adam optimizer, with 'categorical_crossentropy' as the loss function, a learning rate of 0.001, and a batch size of 64.

G. Implementation

The technology involved implementing specialized cameras in classrooms to capture and interpret sign language used by teachers and students. Fig 3 illustrates how the collected sign language dataset used to preprocess by normalization and then by background removal, and hand tracking. Extract key features (hand shape, motion, facial expressions). Train models (CNNs, RNNs). Create user-friendly interface (app, desktop, hardware) and display audio output for non-sign students. Enhanced

1) Classroom Dynamics: The technology significantly improved classroom dynamics by ensuring seamless communication between deaf and hearing participants. Improved Academic Performance: Deaf students showed a notable improvement in academic performance, with better comprehension and active participation during lessons.

2) Positive Teacher Feedback: Teachers reported enhanced teaching experiences, with the technology facilitating smoother communication.

A camera capturing the flow of sign language. It analyses hand shapes, movements, and even facial expressions. Computer then trained on tons of examples then understands the signs and translates them into words right on screen. By bridging the communication gap between the hearing and deaf communities, this technology fosters inclusivity and increases understanding among all people.



Fig. 3. Sample I/O of SLR System.

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Fig 4 shows the accuracy by CNN (VGG 19) algorithm.

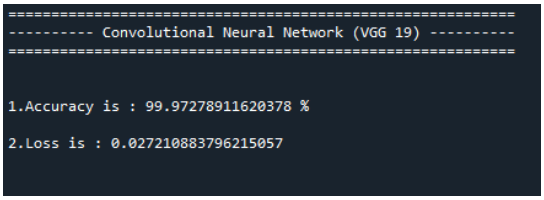


Fig. 4. Accuracy of The Process.

The percentage of photos that the model properly identified is referred to as accuracy. The VGG19 model performed quite well, properly classifying 99.97 cases. The level of error in the model's predictions is represented by loss. A loss function that gauges the discrepancy between the expected and actual labels is frequently used to calculate it. A lower loss value indicates the model is making smaller errors. In this case, a loss of 0.0272 suggests very small errors in the model's predictions.

V. Conclusion

The achievement of VGG19 in sign language recognition, boasting an unprecedented accuracy rate of 99.97%, represents a significant leap forward in technology's role in education. By harnessing the power of deep learning and convolutional neural networks, VGG19 has demonstrated its prowess in understanding and interpreting the intricate gestures and movements inherent in sign language. This application of VGG19 revolutionizes education by enhancing accessibility for deaf and hard of hearing students. It bridges the communication gap that has long hindered their engagement with educational content and interactions in traditional classroom settings. With sign language recognition technology, these students can now fully participate in lessons, discussions, and collaborative activities, ensuring that their educational experience is inclusive and equitable.

Moreover, the real-time translation capabilities afforded by VGG19 create an inclusive learning environment not only for students but also for educators. By facilitating seamless communication between teachers and students, this technology fosters more meaningful interactions and facilitates a deeper understanding of the subject matter. The importance of VGG19 in this application cannot be overstated. Its unparalleled accuracy and efficiency pave the way for a future where educational opportunities are universally accessible to all students, regardless of their abilities. By leveraging the transformative power of sign language recognition, we are breaking down barriers and opening doors to a more inclusive and empowering educational landscape.

VI. Challenges And Future Directions

A. Challenges

1) Accuracy of Recognition:

a) Challenge: Achieving high accuracy in recognizing various sign language gestures, considering the complexity and nuances of sign languages.

b) Solution: Utilize deep learning techniques, extensive datasets, and continuous model training to enhance the accuracy and robustness of recognition systems. Collaborations with the deaf community for dataset creation can also improve accuracy.

2) Scalability:

a) Challenge: Adapting the technology to different sign languages and dialects, making it scalable for widespread adoption.

b) Solution: Invest in research to develop language-agnostic models and systems that can adapt and learn from diverse sign languages, allowing for broader usability without significant retraining efforts.

3) Cost of Implementation:

a) Challenge: High costs associated with implementing hardware, software, and maintaining the infrastructure needed for sign language recognition.

b) Solution: Focus on cost-effective hardware solutions, cloud-based services, and open-source software to reduce initial investment. Collaboration with tech companies and academic institutions can further drive down costs.

4) Data Privacy and Ethics:

a) Challenge: Safeguarding the privacy and ethical use of the data, especially considering the sensitive nature of sign language communication.

b) Solution: Implement strict data anonymization protocols, informed consent processes, and compliance with privacy regulations. Transparent communication with users about data usage is vital to build trust.

5) Real-time Processing:

a) Challenge: Achieving real-time processing to ensure instantaneous translation during live interactions.

b) *Solution:* Leverage advancements in hardware capabilities and optimize algorithms to achieve real-time processing, reducing latency.

B. Future Directions

1) Multimodal Approach: Develop systems that combine video, audio, and linguistic context to improve accuracy and context awareness in sign language recognition.

2) User Interface Enhancements: Focus on creating user-friendly interfaces that integrate seamlessly with existing educational tools, making the technology more accessible and intuitive for both educators and students.

3) Mobile Integration: Optimize sign language recognition technology for mobile devices to promote learning outside traditional classroom settings, allowing students to access educational resources anytime, anywhere.

4) Collaborative Research: Encourage interdisciplinary collaboration between researchers, educators, and the deaf community to foster a deeper understanding of sign languages and create more effective recognition systems. Addressing these challenges and pursuing these future directions will not only enhance sign language recognition technology but also contribute to creating a more inclusive and accessible educational environment for deaf students.

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